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### Gate device, authentication system, gate device control method, and storage medium

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#### Abstract

Provided is a gate device that accurately determines a user who is an authentication target. This gate device is provided with an acquisition unit, a setting unit, and a gate control unit. The acquisition unit acquires an image from a camera. When a plurality of users are included in the acquired image, the setting unit specifies a user on the front row among the plurality of users in accordance with the respective positions of the plurality of users in the image, and sets the specified user as an authentication target person. The gate control unit controls a gate on the basis of a authentication result of the authentication target person.

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## Background/Summary

(1) This application is a Continuation of U.S. application Ser. No. 17/801,664 filed on Aug. 23, 2022, which is a National Stage Entry of PCT/JP2020/012044 filed on Mar. 18, 2020, the contents of all of which are incorporated herein by reference, in their entirety.

### TECHNICAL FIELD

(1) The present disclosure relates to a gate device, an authentication system, a gate device control method, and a storage medium.

### BACKGROUND ART

(2) Services using face authentication have started to spread (see PTLs 1 to 3). For example, face authentication has started to be applied to various procedures (for example, check-in, baggage check, security check, and the like) at airports. A service using face authentication is often provided using a gate device.

### CITATION LIST

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(3) [PTL 1] JP 2006-236244 A [PTL 2] JP 2018-128970 A [PTL 3] JP 2018-109935 A

### SUMMARY

#### Technical Problem

(4) As described above, services using face authentication have started to be provided at airports and the like. For example, a security check using face authentication is performed at a security check site at an airport. Here, the device installation space of the existing airport is limited, and the gate device for face authentication is often installed in a narrow space.

(5) Therefore, when the gate device acquires the image (the image including the face area), the image may include not only the face image of the user in the foremost row (the user of the

authentication target) but also the face image of the user behind. That is, since the gate device supporting the face authentication is installed in a narrow space, the distance between the users aligned in the gate device is short, and a plurality of persons may appear in the same image.

(6) In this case, the gate device needs to select one face image from the plurality of face images and set the selected face image as the authentication target. For example, it is conceivable that the gate device calculates an area (size) for each of a plurality of face images and sets a face image having the largest area as an authentication target. The handling is based on the premise that the face image of the user aligned in the foremost row of the gate device is larger than the face image of the user aligned behind. However, the size of the face has a large individual difference, and in the above correspondence (determination of the authentication target based on the area of the face image), the user aligned behind the user aligned in the foremost row may be set as the authentication target.

(7) A main object of the present disclosure is to provide a gate device, an authentication system, a gate device control method, and a storage medium that contribute to accurately determining a user of the authentication target.

#### Solution to Problem

(8) According to a first aspect of the present disclosure, there is provided a gate device including: an acquisition unit configured to acquire an image from a camera; a setting unit configured, in a case where a plurality of users appears in the acquired image, to specify a user in a foremost row among the plurality of users according to positions of the plurality of users in the image, and set the specified user as an authentication target person; and a gate control unit configured to control a gate based on an authentication result of the authentication target person.

(9) According to a second aspect of the present disclosure, there is provided an authentication system including: a server device configured to store biometric information of each of a plurality of system users and execute authentication processing using the plurality of pieces of biometric information; and a gate device connected to the server device. The gate device includes: an acquisition unit configured to acquire an image from a camera; a setting unit that configured, in a case where a plurality of users appears in the acquired image, to specify a user in a foremost row among the plurality of users according to positions of the plurality of users in the image, and set the specified user as an authentication target person; an authentication request unit configured to request a server device to authenticate the authentication target person; and a gate control unit configured to control a gate based on an authentication result of the authentication target person.

(10) According to a third aspect of the present disclosure, there is provided a gate device control method including: in a gate device, acquiring an image from a camera; in a case where a plurality of users appears in the acquired image, specifying a user in a foremost row among the plurality of users according to positions of the plurality of users in the image, and setting the specified user as an authentication target person; and controlling a gate based on an authentication result of the authentication target person.

(11) According to a fourth aspect of the present disclosure, there is provided a computer-readable storage medium storing a program for causing a computer mounted in a gate device to execute: acquiring an image from a camera; in a case where a plurality of users appears in the acquired image, specifying a user in a foremost row among the plurality of users according to positions of the plurality of users in the image, and setting the specified user as an authentication target person; and controlling a gate based on an authentication result of the authentication target person.

#### Advantageous Effects

(12) According to each aspect of the present disclosure, a gate device, an authentication system, a gate device control method, and a storage medium that contribute to accurately determining a user of the authentication target are provided. Note that the effect of the present disclosure is not limited to the above. According to the present disclosure, other effects may be exhibited instead of or in addition to the effects.

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# Description

## BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a diagram for explaining an outline of an example embodiment.
- (2) FIG. 2 is a diagram illustrating an example of a schematic configuration of an authentication system according to a first example embodiment.
- (3) FIG. 3 is a diagram illustrating an example of an appearance of a gate device according to the first example embodiment.
- (4) FIG. 4 is a diagram illustrating an example of a plan view of the gate device according to the first example embodiment.
- (5) FIG. 5 is a diagram illustrating an example of a processing configuration of the gate device according to the first example embodiment.
- (6) FIG. 6A is a diagram for explaining the operation of a biometric information acquisition unit when a medium-height user is located in the authentication area, in which an upper image is surrounded by a broken line.
- (7) FIG. 6B is a diagram for explaining the operation of a biometric information acquisition unit when a medium-height user is located in the authentication area, in which a lower image is surrounded by a broken line.
- (8) FIG. 6C is a diagram for explaining the operation of a biometric information acquisition unit, in which a region common to the upper image and the lower image is indicated by a one-dot chain line.
- (9) FIG. 7 is a flowchart illustrating an example of an operation of an authentication target setting unit according to the first example embodiment.
- (10) FIG. 8A is a diagram for explaining an operation of the authentication target setting unit, in which the face image of the tall authentication target person is included only in the upper image.
- (11) FIG. 8B is a diagram for explaining an operation of the authentication target setting unit, in which the face image of the short authentication target person is included only in the lower image.
- (12) FIG. 9A is a diagram for explaining an operation of the authentication target setting unit, in which an error is set when a plurality of face images is included only in the lower image.
- (13) FIG. 9B is a diagram for explaining an operation of the authentication target setting unit, in which an error is set when a plurality of face images is included only in the upper image.
- (14) FIG. 10A is a diagram for explaining the operation of the authentication target setting unit, in which a face image extracted from either the upper image or the lower image is set as the authentication target.
- (15) FIG. 10B is a diagram for explaining the operation of the authentication target setting unit, in which of a user U1 and a user U2 is walking in front is identified (determined).
- (16) FIG. 11 is a flowchart illustrating an example of an anteroposterior relationship specifying processing of the authentication target setting unit according to the first example embodiment.
- (17) FIG. 12 is a diagram for explaining the operation of the authentication target setting unit according to the first example embodiment.
- (18) FIG. 13 is a diagram for explaining the operation of the authentication target setting unit according to the first example embodiment.
- (19) FIG. 14 is a diagram illustrating an example of a graph illustrating a correspondence relationship between a distance between a camera installation surface and an object and a position of the object in a common region.
- (20) FIG. 15 is a diagram illustrating an example of an authentication request.
- (21) FIG. 16 is a flowchart illustrating an example of an operation of the gate device according to the first example embodiment.
- (22) FIG. 17 is a diagram illustrating an example of a processing configuration of a server device

according to the first example embodiment.

(23) FIG. **18** is a diagram illustrating an example of a user database.

(24) FIG. **19** is a sequence diagram illustrating an example of an operation of the authentication system according to the first example embodiment.

(25) FIG. **20A** is a diagram for explaining the relationship between the actual position of a user and the position of the user appearing in the image, indicating a case where the user moves to positions **P1**, **P2**, and **P3** is considered.

(26) FIG. **20B** is a diagram for explaining the relationship between the actual position of a user and the position of the user appearing in the image, in which the position of the user in the captured image moves from left to right.

(27) FIG. **21** is a flowchart illustrating an example of an operation of an authentication target setting unit according to a second example embodiment.

(28) FIG. **22** is a diagram illustrating an example of a hardware configuration of the gate device.

#### EXAMPLE EMBODIMENT

(29) First, an outline of an example embodiment will be described. Note that the reference numerals in the drawings attached to this outline are attached to each element for convenience as an example for assisting understanding, and the description of this outline is not intended to be any limitation. In addition, in a case where there is no particular explanation, the block described in each drawing represents not a configuration of a hardware unit but a configuration of a functional unit. Connection lines between blocks in each drawing include both bidirectional and unidirectional lines. The unidirectional arrow schematically indicates a flow of a main signal (data), and does not exclude bidirectionality. Note that, in the present specification and the drawings, elements that can be similarly described are denoted by the same reference numerals, and redundant description can be omitted.

(30) The gate device **100** according to one example embodiment includes an acquisition unit **101**, a setting unit **102**, and a gate control unit **103** (see FIG. **1**). The acquisition unit **101** acquires an image from a camera. When a plurality of users is included in the acquired image, the setting unit **102** specifies a user on the foremost row among the plurality of users in accordance with the positions of the plurality of users in the image, and sets the specified user as an authentication target person. The gate control unit **103** controls a gate on the basis of an authentication result of the authentication target person.

(31) The gate device **100** acquires an image from the camera. In a case where a plurality of users appears in the image, the gate device **100** specifies an authentication target person. For example, the gate device **100** specifies the user in the foremost row based on the position of the user appearing in the image. For example, the gate device **100** specifies the user appearing at the right end or the left end of the image among the plurality of users as the user in the foremost row.

Whether the user at the right end is determined as the foremost row or the user at the left end is determined as the foremost row is based on the relationship between the traveling direction of the user and the installation position of the camera. Details of the relationship between the traveling direction of the user and the installation position of the camera will be described later. The gate device **100** specifies the user in the foremost row regardless of the size of the face area having a large individual difference, so that the user to be authenticated can be accurately determined.

(32) Hereinafter, specific example embodiments will be described in more detail with reference to the drawings.

#### First Example Embodiment

(33) A first example embodiment will be described in more detail with reference to the drawings.

(34) FIG. **2** is a diagram illustrating an example of a schematic configuration of the authentication system according to the first example embodiment. Referring to FIG. **2**, the authentication system includes a plurality of gate devices **10-1** to **10-3** and a server device **20**.

(35) In the following description, in a case where there is no particular reason to distinguish the

gate devices **10-1** to **10-3**, they are simply referred to as “gate device **10**”. Similarly, for other components, reference numerals on the left side separated by hyphens are used to represent the components.

(36) The configuration illustrated in FIG. 2 is an example and is not intended to limit the number of gate devices **10** or the like. The authentication system may include at least one or more gate devices **10**.

(37) The gate device **10** and the server device **20** are configured to be able to communicate with each other by wired or wireless communication means. The server device **20** may be installed in the same building as the gate device **10**, or may be installed on a network (cloud).

(38) The gate device **10** is, for example, a device installed in an airport or a station. The gate device **10** controls passage of a user. The gate device **10** permits passage of a user (hereinafter, described as an authentication successful person) who has been successfully authenticated by the server device **20**. The gate device **10** does not permit passage of a user who has not been authenticated by the server device **20** or a user who has failed in authentication.

(39) The server device **20** is a device that performs authentication processing. Specifically, server device **20** stores biometric information of a user (system user) who can pass through the gate device **10**. The server device **20** executes authentication processing using the biometric information acquired from the gate device **10** and the biometric information stored in advance, and notifies the gate device **10** of an authentication result (authentication success, authentication failure).

(40) The gate device **10** opens the gate to allow the passing of the authentication successful person when the authentication successful person enters the internal area of the own device (the area surrounded by the main body of the gate device **10**).

(41) Examples of the biometric information of the user include data (feature amount) calculated from physical features unique to an individual such as a face and an iris pattern. Alternatively, the biometric information of the user may be image data such as a face image and an iris image. The biometric information of the user may include the physical feature of the user as information.

(42) Next, details of each device included in the authentication system according to the first example embodiment will be described. In the following description, a “face image” of the user will be described as an example of the biometric information.

(43) [Gate Device]

(44) FIG. 3 is a diagram illustrating an example of an appearance of the gate device **10** according to the first example embodiment.

(45) Referring to FIG. 3, the gate device **10** includes a face authentication module **11**. The face authentication module **11** is attached to the main body of the gate device **10**. The face authentication module **11** includes two cameras (upper camera **12-1** and lower camera **12-2**). Note that the “main body” of the gate device **10** is a structure forming the center of the gate device **10**, and is a member that is in contact with the floor and to which a face authentication module and the like are attached.

(46) The face authentication module **11** is installed to be inclined at a predetermined angle with respect to the direction in which the user walks toward the gate device **10**. FIG. 4 is a diagram (plan view) illustrating a case where the gate device **10** is viewed from above.

(47) As illustrated in FIG. 4, the face authentication module **11** is installed to be inclined at an angle  $\theta$  with respect to the traveling direction of the user (the direction perpendicular to a gate **17**). Note that the angle  $\theta$  is selected (adjusted) so that the upper camera **12-1** and the lower camera **12-2** can image a user U in the authentication area.

(48) When the angle  $\theta$  is reduced, only the vicinity of the gate device **10** can be imaged, and there is a possibility that the user enters the inside (internal area) of the gate device **10** before the authentication processing using the face image is ended. In this case, the user needs to stop and stand by until the authentication processing is completed, and it cannot be said that comfortable walk-through can be achieved. On the other hand, when the angle  $\theta$  is increased, it is possible to

capture an image far away from the gate device **10**. In this case, although it is advantageous for achieving a comfortable walk-through, there is a possibility that even a user who is not walking toward the own device may be authenticated. For example, in the example of FIG. **2**, the gate device **10-1** captures an image of a user who is not walking toward the gate device **10-1**, and requests the server device **20** for authentication. As described above, when the angle  $\theta$  is too large, there is a possibility that useless authentication processing occurs. It is desirable that the administrator and the operator of the system determine the optimum angle in consideration of the advantages and disadvantages caused by the setting angle  $\theta$  of the face authentication module **11** as described above. As a result of intensive studies by the inventors, it has been found that the angle  $\theta$  is desirably selected from a range of 30 to 60 degrees.

(49) Although FIGS. **3** and **4** illustrate an example in which the face authentication module **11** is installed on the right side with respect to the traveling direction of the user, it is a matter of course that the installation position of the face authentication module **11** is not limited to the right side. The face authentication module **11** may be installed on the left side with respect to the traveling direction of the user. Although FIGS. **3** and **4** illustrate an example in which the face authentication module **11** is directly installed in the main body of the gate device **10**, the face authentication module **11** may be installed away from the main body of the gate device **10**. Alternatively, the gate device **10** may be mounted on the ceiling portion (roof; not illustrated), the face authentication module **11** may be installed on the ceiling. The face authentication module **11** may be installed at a predetermined angle with respect to the traveling direction of the user.

(50) The upper camera **12-1** is attached to the vertically upper side of the face authentication module **11**. The lower camera **12-2** is attached to the vertically lower side of the face authentication module **11**. That is, as illustrated in FIG. **3**, the lower camera **12-2** is installed on the vertically lower side of the upper camera **12-1**.

(51) On the surface (camera installation surface) of the face authentication module, the positions of the two cameras in the perpendicular direction (vertical direction) are different as described above, but are attached at the same position in the horizontal direction (horizontal position). Similarly, in the depth direction of the two cameras, the two cameras are mounted on the same surface (camera installation surface). Note that the meaning of “the positions of the two cameras in the horizontal direction are the same” does not mean that the positions of the two cameras in the horizontal direction strictly coincide with each other, but allows some deviation. That is, the positions of the two cameras in the horizontal direction may be substantially the same.

(52) The upper camera **12-1** and the lower camera **12-2** are cameras having substantially the same performance, and parameters such as an angle of view and a focal length are substantially the same. According to the method of installing the two cameras or the like as described above, the images obtained from the two cameras have different photographing ranges in the height direction, but have substantially the same photographing ranges in the horizontal direction.

(53) Two cameras (upper camera **12-1** and lower camera **12-2**) included in the gate device **10** are provided to authenticate every person (subject) from a short height to a high height. The upper camera **12-1** is installed to acquire a face image of a tall authentication target person. The lower camera **12-2** is installed to acquire a face image of a short authentication target person.

(54) With these cameras, the face image of the tall authentication target person is acquired by the upper camera **12-1**, and the face image of the short authentication target person is acquired by the lower camera **12-2**. The face image of a medium-height authentication target person is acquired from both the upper camera **12-1** and the lower camera **12-2**. Most of the users are classified as medium height, and it is rare that a face image is acquired from one of the upper camera **12-1** and the lower camera **12-2**.

(55) The upper camera **12-1** and the lower camera **12-2** operate in conjunction with each other. Specifically, the two cameras image the user at substantially the same timing.

(56) As illustrated in FIG. **3**, the face authentication module **11** includes a light emitting diode

(LED) (not illustrated) for dimming and a display **13**. The gate device **10** controls an environment (light irradiated to the user) at the time of acquiring the user's face image using the LED for dimming. The gate device **10** notifies the user of a necessary message or the like by using display **13**. For example, the gate device **10** displays an authentication result (authentication success, authentication failure) by the server device **20**.

(57) The gate device **10** includes a display **14**. The gate device **10** notifies the user of a necessary message or the like by using the display **14**. For example, the gate device **10** displays an authentication result (authentication success, authentication failure) by the server device **20**.

Alternatively, the gate device **10** may notify the user of a necessary message by using a speaker (not illustrated).

(58) The gate device **10** includes an area sensor **15**. The area sensor **15** is a sensor that detects whether a person is present in a predetermined area (an area surrounded by a dotted line in FIG. **3**) set in front of the gate device **10**. The area sensor **15** can be configured using a sensor (so-called human sensor) using infrared rays.

(59) When confirming that a person is present in the predetermined area based on the detection signal from the area sensor **15**, the gate device **10** controls two cameras (upper camera **12-1** and lower camera **12-2**) to acquire two images (image data). In the following description, an area where the area sensor **15** detects the presence of a person is referred to as an "authentication area". When detecting that the user is present in the authentication area, the gate device **10** captures an image of the user located in the authentication area.

(60) The gate device **10** includes an in-gate sensor **16**. The in-gate sensor **16** is a sensor for detecting that a person enters the inside of the own device (gate device **10**). As the in-gate sensor **16**, for example, a sensor (so-called passage sensor using light) including a light transmitting device and a light receiving device can be used. For example, the light transmitting device and the light receiving device are installed to face each other (two devices are installed on an inner wall of the main body). The transmitting device constantly transmits light, and the receiving device receives the transmitted light. The gate device **10** determines that a person enters the gate device **10** when the receiving device cannot receive the light. Note that FIG. **3** illustrates one of the two devices constituting the in-gate sensor **16**. A plurality of pairs of in-gate sensors **16** may be provided at substantially equal intervals in the traveling direction of the person.

(61) The gate device **10** includes the gate **17**. When confirming that the user is present in the authentication area, the gate device **10** acquires a face image of the user. The gate device **10** generates a feature amount (biometric information) from the acquired image, and transmits an authentication request including the generated feature amount to the server device **20**. The gate device **10** opens the gate **17** when the authentication successful person enters the internal area of the own device. The gate device **10** closes the gate **17** after the authentication successful person passes through the gate **17**.

(62) The type of the gate **17** is not particularly limited, and is, for example, a flapper gate in which a flapper provided from one side or both sides of the passage opens and closes, a turnstile gate in which three bars rotate, or the like.

(63) FIG. **5** is a diagram illustrating an example of a processing configuration (processing module) of the gate device **10** according to the first example embodiment. Referring to FIG. **5**, the gate device **10** includes a communication control unit **201**, a user detection unit **202**, a biometric information acquisition unit **203**, an authentication target setting unit **204**, an authentication request unit **205**, an entrant determination unit **206**, a gate control unit **207**, a message output unit **208**, and a storage unit **209**.

(64) The communication control unit **201** is a unit that controls communication with other devices. Specifically, the communication control unit **201** receives data (packet) from the server device **20**. In addition, the communication control unit **201** transmits data to the server device **20**. The communication control unit **201** delivers data received from another device to another processing

module. The communication control unit **201** transmits data acquired from another processing module to another device. In this manner, the other processing modules transmit and receive data to and from other devices via the communication control unit **201**.

(65) The user detection unit **202** is a unit that detects a user located in the authentication area. The user detection unit **202** detects a user present in the authentication area on the basis of a detection signal from the area sensor **15**. In a case where the output of the area sensor **15** indicates that “there is a user in the authentication area”, the user detection unit **202** notifies the biometric information acquisition unit **203** of the fact.

(66) The biometric information acquisition unit **203** is a unit that acquires the biometric information of the user present in the authentication area. Upon acquiring the notification related to “there is a user in the authentication area” from the user detection unit **202**, the biometric information acquisition unit **203** controls the two cameras (upper camera **12-1** and lower camera **12-2**) and captures an image of the user located in the authentication area.

(67) The biometric information acquisition unit **203** acquires images from each of the two cameras. As described above, the upper camera **12-1** is attached to the upper side of the face authentication module **11**. Therefore, the image (hereinafter, referred to as an upper image) obtained from the upper camera **12-1** includes the upper side of the authentication area. For example, when a medium-height user is located in the authentication area, an upper image surrounded by a broken line in FIG. **6A** is obtained.

(68) In contrast, the lower camera **12-2** is attached to the lower side of the face authentication module **11**. Therefore, the image (hereinafter, referred to as a lower image) obtained from the lower camera **12-2** includes the lower side of the authentication area. For example, when a medium-height user is located in the authentication area, a lower image surrounded by a broken line in FIG. **6B** is obtained.

(69) The upper image and the lower image include a common region. In the example of FIGS. **6A** to **6C**, a region indicated by a one-dot chain line in FIG. **6C** is a region common to the upper image and the lower image. In the following description, a region common to the upper image and the lower image is referred to as a “common region”.

(70) The biometric information acquisition unit **203** delivers two images (upper image, lower image) captured by the upper camera **12-1** and the lower camera **12-2** to the authentication target setting unit **204**.

(71) The authentication target setting unit **204** is a unit that sets (determines) a user (authentication target person) to be authenticated from users appearing in an image. When there are a plurality of users in the authentication area, the authentication target setting unit **204** selects one user from the plurality of users and sets the selected user as the “authentication target person”. The operation of the authentication target setting unit **204** will be described with reference to the flowchart illustrated in FIG. **7**.

(72) The authentication target setting unit **204** extracts a face image (Step **S101**). The authentication target setting unit **204** extracts a face image from the upper image. The authentication target setting unit **204** extracts a face image from the lower image. The authentication target setting unit **204** extracts a face image from the image using various methods. For example, the authentication target setting unit **204** extracts a face image (face area) from the image data using a learning model learned by a convolutional neural network (CNN). Alternatively, the authentication target setting unit **204** may extract the face image using a method such as template matching.

(73) The authentication target setting unit **204** determines the distribution status of the face images in the two images (Step **S102**). The authentication target setting unit **204** determines whether the face image is included only in the upper image or the lower image or whether the face image is included in both the upper image and the lower image. In the former case, the processing in and after Step **S103** is executed. In the latter case, the processing from Step **S110** is executed.

(74) When Step S102 is executed and it is determined that the face image is distributed in the upper image or the lower image, the authentication target setting unit 204 determines whether there is one face image included only in the upper image or the lower image (Step S103).

(75) When one face image is included in the upper image or the lower image (Step S103: Yes branch), the authentication target setting unit 204 sets the face image as the authentication target (Step S104).

(76) For example, as illustrated in FIG. 8A, the face image of the tall authentication target person is included only in the upper image. In addition, as illustrated in FIG. 8B, the face image of the short authentication target person is included only in the lower image. In the case illustrated in FIGS. 8A and 8B, the authentication target setting unit 204 sets the face image included in the upper image and the face image included in the lower image as the authentication target (the face image of the authentication target person). Note that, in the drawings including FIGS. 8A and 8B, the range of each image is moved in the horizontal direction in order to facilitate distinction between the upper image and the lower image.

(77) When two or more face images are included in the upper image or the lower image (Step S103: No branch), the authentication target setting unit 204 sets an error (Step S105).

(78) For example, as illustrated in FIGS. 9A and 9B, an error is set when a plurality of face images is included only in the upper image or the lower image. In the case illustrated in FIGS. 9A and 9B, it is difficult to determine which one of the plurality of face images is set as the authentication target. However, the situation illustrated in FIGS. 9A and 9B rarely occurs. When such a situation occurs, the gate device 10 takes a measure such as instructing the user to leave the authentication area.

(79) As described above, in a case where the face image is distributed in each of the upper image and the lower image, the processing after Step S110 illustrated in FIG. 7 is executed.

(80) The authentication target setting unit 204 determines whether one face image is included in each of the upper image and the lower image (Step S110).

(81) Specifically, the authentication target setting unit 204 determines whether a situation as illustrated in FIG. 10A has occurred or a situation as illustrated in FIG. 10B has occurred.

(82) When one face image is included in each of the upper image and the lower image (Step S110: Yes branch), the authentication target setting unit 204 sets the face image included in either the upper image or the lower image as the authentication target (Step S111). In the example of FIG. 10A, a face image extracted from either the upper image or the lower image is set as the authentication target.

(83) When two or more face images are included in each of the upper image and the lower image (Step S110: No branch), the authentication target setting unit 204 specifies the anteroposterior relationship of each user (Step S112). In the example of FIG. 10B, which of a user U1 and a user U2 is walking in front is identified (determined). Details regarding the identification processing of the anteroposterior relationship will be described later.

(84) The authentication target setting unit 204 sets a user in the foremost row (forefront) among a plurality of users (a plurality of users present in the authentication area) as the authentication target (Step S113). In the example of FIG. 10B, in a case where the user U1 is walking in front of the user U2 (in a case where the user U1 is closer to the gate device 10 than the user U2), the face image of the user U1 is set as the authentication target.

(85) As described above, when one user appears in both the upper image and the lower image, the authentication target setting unit 204 sets the one user as the authentication target person. In addition, when one user appears in one of the upper image and the lower image, the authentication target setting unit 204 sets the one user as the authentication target person.

(86) Note that the authentication target setting unit 204 desirably confirms whether the users appearing in the upper image and the lower image are the same person before setting the authentication target in Step S111. Specifically, the authentication target setting unit 204 calculates

the feature amount from each of the two face images. The authentication target setting unit **204** determines whether the two feature amounts substantially match. The authentication target setting unit **204** calculates similarity between two feature amounts (feature vectors). If the calculated similarity is equal to or more than a predetermined value, the authentication target setting unit **204** determines that the two feature amounts substantially match. That is, the authentication target setting unit **204** determines whether the two face images are face images of the same person by one-to-one collation using the feature amounts generated from the upper face image and the lower face image. When the two feature amounts do not match, the authentication target setting unit **204** may set an error. That is, a situation in which one face image is included in each of the upper image and the lower image and each face image is not a face image of the same person does not usually occur. Therefore, it is appropriate to set the situation as an error.

(87) In addition, in a case where there are one tall authentication target person and one short authentication target person in the authentication area, the processing related to Step **S112** is executed. However, in this case, since there is no common region between the upper image and the lower image, the anteroposterior relationship in Step **S112** cannot be specified. Therefore, when the above situation occurs, the authentication target setting unit **204** sets an error.

(88) Next, the processing (the identification processing of the anteroposterior relationship) according to Step **S112** in FIG. 7 will be described. FIG. 11 is a flowchart illustrating an example of the anteroposterior relationship specifying processing of the authentication target setting unit **204**.

(89) The authentication target setting unit **204** sets the “distance measurement band” in the common region of the two images (upper image, lower image) (Step **S201**). For example, in a case where the upper image and the lower image illustrated in FIG. 12 are obtained, the region colored in gray is set as the “distance measurement band”. As illustrated in FIG. 12, the distance measurement band is desirably set on the lower side of the upper image. The distance measurement band is set to a region in which two or more users simultaneously appear in the common region. In an extreme example, the distance measurement band may be set at the lowermost side of the upper image, or the entire common region may be set as the distance measurement band. The authentication target setting unit **204** sets all or a part of the upper image and the lower image in the distance measurement band.

(90) The authentication target setting unit **204** calculates the distance to the camera installation surface for the object appearing in the distance measurement band (all or part of the common region common to the upper image and the lower image) (Step **S202**). Specifically, the authentication target setting unit **204** calculates the distance to the camera (upper camera **12-1** and lower camera **12-2**) installation surface for each small area included in the distance measurement band. The authentication target setting unit **204** calculates the distance between each small area (object) and the camera installation surface using the parallax (deviation between upper image and lower image) of the two images.

(91) The authentication target setting unit **204** calculates a deviation amount (parallax) between the upper image and the lower image regarding the object included in the distance measurement band. The authentication target setting unit **204** calculates the distance between the object and the camera installation surface using the calculated parallax. That is, the authentication target setting unit **204** calculates the distance between the camera installation surface and the object (small region of the distance measurement band) using a so-called stereo method. Although the calculation of the distance using the stereo method is obvious to those skilled in the art and detailed description is omitted, the authentication target setting unit **204** calculates the distance between the camera installation surface and the object by the following method.

(92) The authentication target setting unit **204** divides the distance measurement band into a plurality of small areas. For example, the authentication target setting unit **204** divides the distance measurement band such that the size of one area is  $A1 \times A2$  pixels. The authentication target setting

unit **204** calculates a distance D between the small area (object) and the camera installation surface by the following Expression (1).

(93) [Math. 1]  $D = B \times f / Z$  (1)

(94) In Expression (1), B is a distance (camera interval) between the upper camera **12-1** and the lower camera **12-2**. f is the focal length of the two cameras. Z is parallax (deviation amount). FIG. **13** illustrates the relationship between the parameters expressed by Expression (1).

(95) In Expression (1), the camera interval B and the focal length f are values determined in advance according to the installation and selection of the two cameras. Therefore, the authentication target setting unit **204** can calculate the distance D by obtaining the parallax Z.

(96) The authentication target setting unit **204** searches for a region having a matching pattern in a small region forming a common region band of the reference camera (the upper camera **12-1**) with respect to one small region in the image (distance measurement band) of the camera (the lower camera **12-2** in FIG. **13**) as a base. For example, the authentication target setting unit **204** calculates a city block distance for each small region, and selects a region having the shortest distance as a region with the same pattern. The authentication target setting unit **204** calculates a deviation between the small region on the base camera side and the small region on the reference camera side selected as described above as the parallax Z.

(97) For example, when the distance D between the object included in the distance measurement band illustrated in FIG. **12** and the camera installation surface is calculated, and the position (position in the horizontal direction) of the object is set to the X axis and the calculated distance D is set to the Y axis, a graph as illustrated in FIG. **14** is obtained.

(98) In order to obtain a waveform as illustrated in FIG. **14**, the authentication target setting unit **204** desirably performs predetermined data processing or the like on the calculated distance D. For example, in order to remove a high frequency component (a high frequency component of the distance D) caused by noise included in the image, the authentication target setting unit **204** may apply a low pass filter (LPF) to the calculated distance D. Alternatively, in a case where the variation of the distance D falls within a predetermined range, the authentication target setting unit **204** may execute processing of converging each value of the distance D to a predetermined value (for example, the average value of the variation values).

(99) As illustrated in FIG. **14**, when a plurality of users is imaged by the upper camera **12-1** and the lower camera **12-2**, peaks depending on the number of people appearing in the two images appear. In the examples of FIGS. **12** and **14**, two peaks appear.

(100) The authentication target setting unit **204** specifies a position (X coordinate) where each peak appears (Step S203). In the example of FIG. **14**, coordinates X1 and X2 are specified. That is, the authentication target setting unit **204** specifies (searches) peaks depending on the number of the plurality of users in the position of the object (the position of the object in the image) and the distance (for example, a graph of FIG. **14** illustrating the relationship between the position and the distance.) to the camera installation surface of the object.

(101) The authentication target setting unit **204** specifies the peak of the distance D using an arbitrary method. For example, the authentication target setting unit **204** calculates a difference between data regarding the distance D, and sets a point at which the calculated difference changes from positive to negative as a peak. It is desirable that the authentication target setting unit **204** specify the peak from the data to which the filtering processing and the convergence processing are applied.

(102) The authentication target setting unit **204** selects the X coordinate related to the minimum peak value (Step S204). The user (face image) located on the vertically upper side of the selected X coordinate is specified as a person in the foremost row of the authentication area. In the examples of FIGS. **12** and **14**, the user U1 located above the coordinate X1 is specified as the user in the foremost row. That is, the authentication target setting unit **204** treats the user at the position related

to the minimum peak value among the plurality of specified peaks as the user in the foremost row.

(103) The authentication target setting unit **204** sets the face image of the user in the foremost row as the authentication target (Step **S113** in FIG. 7). As described above, in a case where a plurality of users appears in both the upper image and the lower image, the authentication target setting unit **204** specifies the user in the foremost row from among the plurality of users by calculating the distance between each of the plurality of users and the camera installation surface. The authentication target setting unit **204** sets the specified user as an authentication target person.

(104) The authentication target setting unit **204** delivers the face image (biometric information) of the authentication target person to the authentication request unit **205**.

(105) The authentication request unit **205** is a unit that requests the server device **20** to authenticate the authentication target person. The authentication request unit **205** generates a feature amount (a feature vector including a plurality of feature amounts) from the face image acquired from the biometric information acquisition unit **203**. The authentication request unit **205** transmits an authentication request including the generated feature amount (biometric information) to the server device **20**.

(106) Note that an existing technique can be used for the feature amount generation processing by the authentication request unit **205**, and thus, a detailed description thereof will be omitted. For example, the authentication request unit **205** extracts eyes, a nose, a mouth, and the like as feature points from the face image. Thereafter, the authentication request unit **205** calculates the position of each feature point and the distance between the feature points as feature amounts, and generates a feature vector (vector information characterizing the face image) including a plurality of feature amounts.

(107) The authentication request unit **205** generates an authentication request including an identifier (hereinafter, referred to as a gate identifier) of the own device, a feature amount, and the like (see FIG. 15). As the gate identifier, a media access control (MAC) address or an Internet protocol (IP) address of the gate device **10** can be used.

(108) When transmitting the authentication request to the server device **20**, the authentication request unit **205** delivers the feature amount included in the request to the entrant determination unit **206**.

(109) The authentication request unit **205** receives a response to the authentication request from the server device **20**. The authentication request unit **205** delivers a response (authentication success, authentication failure) from the server device **20** to the gate control unit **207**.

(110) The entrant determination unit **206** is a unit that detects an entrant using the in-gate sensor **16** for detecting an entrant into the own device (gate device **10**) and determines whether the entrant and the authentication target person are the same person. The entrant determination unit **206** monitors (polls) the output of the in-gate sensor **16**. When the output of the in-gate sensor **16** indicates “there is an entrant inside”, the entrant determination unit **206** determines whether the entrant and the authentication target person match. Specifically, similarly to the biometric information acquisition unit **203**, the entrant determination unit **206** controls at least one camera to acquire a face image. The entrant determination unit **206** calculates a feature amount (a feature amount of an entrant) from the obtained face image.

(111) The entrant determination unit **206** determines whether the calculated feature amount (the feature amount of the entrant) and the feature amount (the feature amount of the authentication target person) acquired from the authentication request unit **205** substantially match. Specifically, the entrant determination unit **206** calculates the similarity between the two feature amounts. A chi-square distance, a Euclidean distance, or the like can be used for the similarity. Note that the similarity is lower as the distance is longer, and the similarity is higher as the distance is shorter. When the similarity is equal to or greater than a predetermined value, the entrant determination unit **206** determines that the two feature amounts substantially match (the entrant and the authentication target person are the same person).

(112) In this manner, the entrant determination unit **206** detects an entrant based on the detection signal from a passage sensor (the in-gate sensor **16**) using light. Thereafter, the entrant determination unit **206** performs one-to-one collation using two feature amounts (feature amount of entry person, feature amount of authentication target person). Whether the entrant and the authentication target person are the same person is determined by the one-to-one collation (one-to-one collation using the biometric information of the entrant and the biometric information of the authentication target person).

(113) When determining that the entrant and the authentication target person are the same person, the entrant determination unit **206** notifies the gate control unit **207** of the fact.

(114) The gate control unit **207** is a unit that controls the gate **17** included in the gate device **10**. The gate control unit **207** controls the gate **17** based on the authentication result of the authentication target person. The gate control unit **207** opens the gate **17** when the authentication result of the server device **20** is “authentication success” and the entrant and the authentication target person are the same person. In other words, in principle, the gate control unit **207** does not open the gate **17** unless the above condition (authentication success, the entrant and the authentication target person are the same person) is satisfied. The gate control unit **207** closes the gate **17** after a user permitted to pass by using a distance sensor or the like passes through the gate **17**.

(115) The message output unit **208** is a unit that outputs a message to be notified to the user (authentication target person, authentication successful person, etc.). The message output unit **208** notifies the user of a necessary message using the displays **13** and **14**, a speaker, or the like.

(116) The storage unit **209** is a unit that stores information necessary for the operation of the gate device **10**.

(117) FIG. **16** is a flowchart illustrating an example of the operation of the gate device **10** according to the first example embodiment.

(118) The gate device **10** detects whether the user is present in the authentication area (Step **S301**). When the user is not detected in the authentication area (Step **S301**: No branch), the gate device **10** repeats the process in Step **S301**.

(119) When the user is detected (Step **S301**: Yes branch), the gate device **10** images the user located in the authentication area and acquires the face image (Step **S302**).

(120) The gate device **10** calculates a feature amount from the face image (Step **S303**).

(121) The gate device **10** transmits an authentication request including the calculated feature amount to the server device **20**, and receives a response (authentication result) to the authentication request (Step **S304**).

(122) The gate device **10** determines whether an entrant is present in the internal area of the own device (Step **S305**). When the entrant cannot be detected even after the lapse of a predetermined time from the start of the process of detecting the entrant (Step **S305**: No branch), the gate device **10** executes the processing related to Step **S309**.

(123) When the entrant is detected during the predetermined period (Step **S305**: Yes branch), the gate device **10** determines whether the entrant and the authentication target person are the same person (Step **S306**). When the entrant and the authentication target person are not the same person (Step **S306**: No branch), the gate device **10** executes the processing related to Step **S309**.

(124) If the entrant and the authentication target person are the same person (Step **S306**: Yes branch), the gate device **10** determines whether the authentication result acquired from the server device **20** is “authentication success” (Step **S307**). When the authentication result is “authentication failure”, the gate device **10** executes the processing related to Step **S309**. Note that it is a matter of course that the processing related to Step **S307** may be executed before the processing related to Step **S305**.

(125) If the authentication result is “authentication success”, the gate device opens the gate **17** (Step **S308**). When a gate opening condition (the authentication success, and the entrant and the

authentication target person are the same person) is satisfied, the gate device **10** controls the gate **17** so that the authentication target person can pass through. At that time, the gate device **10** may display on displays **13** and **14** that the authentication of the authentication target person has succeeded.

(126) When the above condition is not satisfied (No branch in any of Steps **S305**, **S306**, and **S307**), the gate device **10** performs error processing (Step **S309**). Specifically, when the entrant cannot be detected during the predetermined period (Step **S305**: No branch), the gate device **10** discards the authentication result acquired from the server device **20**. When the entrant and the authentication target person are different (Step **S306**: No branch), the gate device **10** displays the fact on the displays **13** and **14**. When the authentication by the server device **20** fails (Step **S307**: No branch), the gate device **10** displays the fact on the displays **13** and **14**.

(127) In this way, the gate device **10** acquires the biometric information (face image) on the users from the short height to the high height using the two cameras (upper camera **12-1** and lower camera **12-2**) installed vertically. In addition, the gate device **10** uses the parallax of the object commonly appearing in the images obtained from the two cameras, determines the anteroposterior relationship in a case where a plurality of users is imaged, and specifies the correct authentication target person. That is, the gate device **10** uses the two cameras to acquire biometric information from users having greatly different heights without omission, and prevents erroneous determination of the authentication target person.

(128) [Server Device]

(129) FIG. **17** is a diagram illustrating an example of a processing configuration (processing module) of the server device **20** according to the first example embodiment. Referring to FIG. **17**, the server device **20** includes a communication control unit **301**, a user registration unit **302**, an authentication unit **303**, and a storage unit **304**.

(130) The communication control unit **301** is a unit that controls communication with other devices. Specifically, the communication control unit **301** receives data (packet) from the gate device **10**. Further, the communication control unit **301** transmits data to the gate device **10**. The communication control unit **301** delivers data received from another device to another processing module. The communication control unit **301** transmits data acquired from another processing module to another device. In this manner, the other processing modules transmit and receive data to and from other devices via the communication control unit **301**.

(131) The user registration unit **302** is a unit that performs system registration of a user (system user) who can pass through the gate device **10**. The user registration unit **302** acquires biometric information (for example, a face image) and personal information (for example, the name) of a user who can pass through the gate device **10** using an arbitrary means.

(132) For example, in a case where the gate device **10** is a device installed at an airport, the system user inputs biometric information and personal information (name, passport number, etc.) to the server device **20** using a web page of an airline company or a kiosk terminal installed at the airport. Alternatively, in a case where the gate device **10** is a device installed at a ticket gate of a station, the system user inputs biometric information, personal information, and the like to the server device **20** from a web page of a railway company or the like or a terminal installed at a station.

(133) When acquiring a face image, the user registration unit **302** calculates a feature amount from the face image. The user registration unit **302** registers the biometric information (for example, the feature amount calculated from the face image) of the system user in the “user database”. The user registration unit **302** registers a user identifier (ID) for identifying the system user, personal information (for example, name, nationality, gender, and the like), and biometric information in the user database in association with each other as necessary (see FIG. **18**). The user database illustrated in FIG. **18** is an example, and other items may be stored in association with the biometric information (feature amount). For example, a “face image” may be registered in the user database.

(134) The authentication unit **303** is a unit that processes the authentication request received from the gate device **10**. Specifically, the authentication unit **303** sets the biometric information (feature amount) included in the authentication request as the collation target, and performs the collation processing with the biometric information registered in the user database.

(135) More specifically, the authentication unit **303** sets the feature amount extracted from the authentication request as the collation target, and executes one-to-N (N is a positive integer, and the same applies hereinafter) collation with the plurality of feature amounts registered in the user database.

(136) The authentication unit **303** calculates similarity between the feature amount (feature vector) of the collation target and each of the plurality of feature amounts on the registration side. The authentication unit **303** sets “authentication success” if the feature amount of which the similarity is equal to or greater than the predetermined value is registered in the user database. The authentication unit **303** sets “authentication failure” if the feature amount of which the similarity is equal to or greater than the predetermined value is not registered in the user database.

(137) In addition, the authentication unit **303** specifies a feature amount of which the similarity with the feature amount of the collation target is equal to or greater than a predetermined value and of which the similarity is the highest among the plurality of feature amounts registered in the user database. The authentication unit **303** transmits the user ID and the personal information related to the specified feature amount to another module and the gate device **10** as necessary. For example, in a case where the gate device **10** is a device installed in an airport, the progress (check passage, security check passage, etc.) of the procedure is managed using the information (name, passport number, etc.) specified by the authentication unit **303**. For example, in a case where gate device **10** is a device installed at a ticket gate of a station, fare adjustment processing or the like is performed using the specified information.

(138) The storage unit **304** stores various types of information necessary for the operation of the server device **20**. In the storage unit **304**, a user database is constructed.

(139) FIG. **19** is a sequence diagram illustrating an example of the operation of the authentication system according to the first example embodiment. It is assumed that the system user is registered in advance prior to the operation of FIG. **19**.

(140) The gate device **10** detects the user located in the authentication area (Step **S01**). Note that the gate device **10** may take a measure such that the user can recognize that the authentication processing is being executed when the authentication target person is detected. For example, the gate device **10** may display the face image of the authentication target person on the displays **13** and **14**. Alternatively, the gate device **10** may display, on displays **13** and **14**, a message or a symbol (for example, an arrow or the like) prompting the user to enter the internal area of the gate device **10** from the authentication area.

(141) The gate device **10** acquires the biometric information of the authentication target person (the user located in the authentication area), and transmits an authentication request including the biometric information to the server device **20** (Step **S02**).

(142) The server device **20** executes authentication processing using the biometric information included in the authentication request and the biometric information registered in the user database (Step **S11**).

(143) The server device **20** transmits the authentication result (authentication success, authentication failure) to the gate device **10** (Step **S12**).

(144) The gate device **10** determines whether the authentication target person enters in parallel with the authentication processing of the server device **20** (detection of the entrant; Step **S03**).

(145) The gate device **10** checks whether a condition for opening the gate **17** (gate opening condition) is satisfied (Step **S04**). Specifically, the gate device confirms whether the authentication succeeds and whether the entrant and the authentication target person are the same person.

(146) When the gate opening condition is satisfied, the gate device **10** opens the gate **17** (Step **S05**).

(147) As described above, the authentication system according to the first example embodiment specifies the authentication target person in a case where a plurality of persons (candidates of system user and authentication target person) is captured in images obtained from two cameras (upper camera **12-1** and lower camera **12-2**). At that time, the gate device **10** calculates the distance between each user and the camera installation surface using the parallax of the object (the user commonly appearing in the two images) in the two images. The gate device **10** specifies a user closest to the own device based on the calculated distance, and sets the specified user as an authentication target person. As described above, in the first example embodiment, the user in the foremost row is specified regardless of the size of the face area having a large individual difference, so that the user to be authenticated can be accurately determined.

#### Second Example Embodiment

(148) Next, a second example embodiment will be described in detail with reference to the drawings.

(149) In the first example embodiment, the distance between the common region of the upper image and the lower image and the camera installation surface is calculated, and the anteroposterior relationship of the user is determined based on the calculated distance. In the second example embodiment, a case where the anteroposterior relationship is determined on the basis of the positional relationship of the user appearing in the image will be described.

(150) Since the configuration of the authentication system according to the second example embodiment can be the same as that of the first example embodiment, the description related to FIG. 2 is omitted. In addition, since the processing configurations of the gate device **10** and the server device **20** according to the second example embodiment can be the same as those of the first example embodiment, the description thereof will be omitted.

(151) In the first example embodiment, the case where the user is imaged using the two cameras has been described. However, in the second example embodiment, it is sufficient that the image of the user can be capture using at least one or more cameras. For example, in a case where a user with a high height and a short height can be imaged by one camera, two cameras are unnecessary. Alternatively, the two cameras are unnecessary even in a case where authentication regarding a user with a high height and a short height is unnecessary or an error is allowed at the time of authentication. It is a matter of course that the gate device **10** according to the second example embodiment may include two cameras. In the following description, a case where the gate device **10** according to the second example embodiment includes two cameras will be described.

(152) Hereinafter, differences between the first and second example embodiments will be mainly described.

(153) As described above, the face authentication module **11** is installed to be inclined at a predetermined angle with respect to the traveling direction of the user. For example, the face authentication module **11** is installed at 45 degrees ( $\theta$  illustrated in FIG. 4 is 45 degrees).

(154) Since the face authentication module **11** is installed to be inclined with respect to the traveling direction of the user, the position where the user appears in the image changes according to the position of the user (distance between the user and the gate device **10**). Specifically, in a case where the user is present far away, the user appears on the side where the face authentication module **11** of the gate device **10** is installed (the left direction in the example of FIG. 4). Thereafter, when the user approaches the gate device **10**, the position of the user (the position of the user shown in the image) moves on the right side (in the rightward direction; the X coordinate in a case where the lower left of the image is set as the origin moves in the positive direction).

(155) For example, as illustrated in FIG. 20A, a case where the user moves to positions P1, P2, and P3 is considered. In this case, the position of the user in the captured image moves from left to right as illustrated in FIG. 20B. This fact indicates that the user appeared on the right side of the image is close to the gate device **10**. In other words, in a case where a plurality of users appears in the same image, it indicates that the user located on the right side of the image is closer to the gate device **10**

than the user located on the left side. The gate device **10** according to the second example embodiment specifies the anteroposterior relationship among the plurality of users using the fact. (156) The authentication target setting unit **204** of the gate device **10** according to the second example embodiment is different from that of the first example embodiment in the processing of Step **S112** illustrated in FIG. **7**. In a case where a plurality of users appears in the image, the authentication target setting unit **204** specifies the user in the foremost row among the plurality of users according to the position of each of the plurality of users in the image. The authentication target setting unit **204** sets the specified user as an authentication target person.

(157) FIG. **21** is a flowchart illustrating an example of the operation of the authentication target setting unit **204** according to the second example embodiment. With reference to FIG. **21**, the anteroposterior relationship specifying processing of the authentication target setting unit will be described.

(158) The authentication target setting unit **204** calculates X coordinates (coordinates in the horizontal direction) of each face image included in the image (Step **S401**). Note that the calculation of the X coordinates may be performed using any one of the upper image and the lower image.

(159) The authentication target setting unit **204** selects the X coordinate of the right end (Step **S402**). The face image related to the selected X coordinate is to be authenticated.

(160) The authentication target setting unit **204** may apply a predetermined condition to a target for specifying the anteroposterior relationship to a plurality of users (face images) included in the image. The authentication target setting unit **204** may set a face image satisfying a predetermined condition as the identification target of the anteroposterior relationship. Specifically, the authentication target setting unit **204** may exclude a user whose interocular distance is smaller than a predetermined value (for example, 60 pixels) from the determination related to the user in the foremost row among the plurality of users. By setting such a condition, it is possible to exclude a user who appears at the right end of the image while being located far from the gate device **10**.

(161) Note that, in the gate device **10** illustrated in FIGS. **3** and **4**, since the face authentication module **11** is installed on the right side with respect to the traveling direction of the user, the user appearing at the right end of the image is set as the authentication target person. In a case where the face authentication module **11** is installed on the left side with respect to the traveling direction of the user, the user appearing at the left end of the image is set as the authentication target person.

(162) As described above, the gate device **10** according to the second example embodiment specifies the user located at the end on the side determined according to the traveling direction of the user and the installation position of the camera as the user in the foremost row. As a result, similarly to the first example embodiment, the gate device **10** can accurately determine the user of the authentication target.

(163) Next, hardware of each device constituting the authentication system will be described. FIG. **22** is a diagram illustrating an example of a hardware configuration of the gate device **10**. Note that, since the face authentication module **11**, the camera **12**, and the like have already been described, these elements are not illustrated in FIG. **22**.

(164) The gate device **10** includes a processor **311**, a memory **312**, an input/output interface **313**, a communication interface **314**, and the like. The components such as the processor **311** are connected by an internal bus or the like and are configured to be able to communicate with each other.

(165) However, the configuration illustrated in FIG. **22** is not intended to limit the hardware configuration of the gate device **10**. The gate device **10** may include hardware (not illustrated) or may not include the input/output interface **313** as necessary. In addition, the number of processors **311** and the like included in the gate device **10** is not limited to the example of FIG. **22**, and for example, a plurality of processors **311** may be included in the gate device **10**.

(166) The processor **311** is a programmable device such as a central processing unit (CPU), a micro

processing unit (MPU), or a digital signal processor (DSP). Alternatively, the processor **311** may be a device such as a field programmable gate array (FPGA) or an application specific integrated circuit (ASIC). The processor **311** is configured to execute various programs including an operating system (OS).

(167) The memory **312** is a random access memory (RAM), a read only memory (ROM), a hard disk drive (HDD), a solid state drive (SSD), or the like. The memory **312** stores an OS program, an application program, and various data.

(168) The input/output interface **313** is an interface of a display device or an input device (not illustrated). The display device may be the display **14**. The input device may be a touch panel integrated with the display **14**.

(169) The communication interface **314** is a circuit, a module, or the like that communicates with another device. For example, the communication interface **314** includes a network interface card (NIC) or the like.

(170) The functions of the gate device **10** are implemented by various processing modules. The processing module is implemented, for example, by the processor **311** executing a program stored in the memory **312**. In addition, the program can be recorded in a computer-readable storage medium. The storage medium may be a non-transient (non-transitory) medium such as a semiconductor memory, a hard disk, a magnetic recording medium, or an optical recording medium. That is, the present disclosure can also be embodied as a computer program product. In addition, the program can be downloaded via a network or updated using a storage medium storing the program. Further, the processing module may be achieved by a semiconductor chip.

(171) The server device **20** can be configured similarly to the gate device **10**, and a basic hardware configuration of the server device is not different from that of the gate device **10**. Therefore, the description is omitted.

(172) The gate device **10** is equipped with a computer, and can achieve the function of the gate device **10** by causing the computer to execute a program. The gate device **10** executes the “gate device control method” by the program.

#### Modified Example

(173) Note that the configuration, operation, and the like of the authentication system described in the above example embodiments are merely examples, and are not intended to limit the configuration and the like of the system.

(174) The application destination of the gate device **10** of the present disclosure is not particularly limited. For example, “airport” is exemplified as an application destination (installation place) of the gate device **10** of the present disclosure. Specifically, the gate device **10** may be used as a part of an apparatus responsible for an examination system using biometric authentication performed at an airport. In the examination system, identity confirmation (confirmation as to whether the photographed face image matches the face image described in the passport) is performed at the time of check-in, and face data (face image or feature amount), flight information, and passport information are stored in association with each other. When the identity confirmation is completed, a token for performing examination in biometric authentication is generated. The gate device **10** of the present disclosure is installed in various examination areas (for example, the security inspection site) in an airport, and the gate device **10** authenticates a user by collating a face image registered at the time of check-in with a face image captured by the own device.

(175) In the above example embodiments, the case of detecting an entrant using the in-gate sensor **16** has been described. However, when it is difficult for a user different from the authentication target person to enter the gate device **10**, the detection of the entrant is unnecessary. For example, by configuring the length of the main body of the gate device **10** in the traveling direction of the person passing through the gate to be long and setting the authentication area inside gate device **10**, it is difficult for a user different from the authentication target person to enter the inside of the gate device **10**.

- (176) In the above example embodiments, the gate device **10** including the two cameras (upper camera **12-1** and lower camera **12-2**) has been described. However, the gate device **10** may include one camera. In this case, one camera may be configured to be movable up and down, and the gate device **10** may acquire two images (upper image, lower image) by changing a position of the camera.
- (177) In the above example embodiments, the case where the user is imaged by the two cameras (upper camera **12-1** and lower camera **12-2**), and the authentication target person (the user in the foremost row) is specified using the parallax obtained from the two images has been described. However, the number of cameras is not limited to two, and the authentication target person may be specified using three or more cameras. For example, a middle camera is installed between the upper camera **12-1** and the lower camera **12-2**. The gate device **10** can determine the anteroposterior relationship between the users appearing in the upper camera **12-1** and the middle camera and the anteroposterior relationship between the users appearing in the middle camera and the lower camera **12-2**.
- (178) In the above example embodiments, the description has been made on the assumption that the performance of the upper camera **12-1** and the lower camera **12-2** is substantially the same. However, the performance of these cameras may be different. In this case, the gate device **10** may take measures to fill the performance difference between the two cameras. For example, the gate device **10** may convert the size or the like of one of the obtained two images. That is, the gate device of the disclosure of the present application can be achieved even in a case where the selection of the camera that can be used is widened by the measures such as the conversion processing, and the installation place and the like of the camera in the face authentication module **11** are limited.
- (179) In the above example embodiments, the case where the biometric information related to the feature amount generated from the face image is transmitted from the gate device **10** to the server device **20** has been described. However, the “face image” itself may be transmitted as the biometric information from the gate device **10** to the server device **20**. The server device may generate a feature amount from the acquired face image and execute authentication processing (one-to-N collation). Even when the face image itself is transmitted to the server device **20**, the gate device **10** determines the identity between the authentication target person and the entrant using the feature amount generated from the face image.
- (180) In the above example embodiments, the determination on the identity between the authentication target person and the entrant is executed by the gate device **10**, but the determination may be executed by the server device **20**. In this case, the gate device **10** acquires biometric information of an entrant into the internal area, and transmits the biometric information to the server device **20**. The server device **20** may collate the biometric information included in the authentication request with the biometric information of the entrant to determine the identity between the authentication target person and the entrant.
- (181) In the above example embodiments, a so-called client server system has been described in which the gate device **10** transmits the authentication request to the server device **20**, and the server device **20** processes the request. However, the authentication processing executed by the server device **20** may be executed by the gate device **10**. That is, some or all of the functions of server device **20** may be implemented by the gate device **10**.
- (182) In the above example embodiments, it has been described that an error is set in a case where the face image is distributed in either the upper image or the lower image and the image includes a plurality of face images (Step **S105** in FIG. 7). In this case, when the upper image includes a plurality of face images, the gate device **10** may determine the anteroposterior relationship using the parallax between the two images and the position of the user. For example, in the case of FIG. **9A**, the anteroposterior relationship may be determined using an object (torso, legs, or the like of the user) that appears in common in the upper image and the lower image. Alternatively, as

described in the second example embodiment, the gate device **10** may determine the anteroposterior relationship according to the position of the face image included in the upper image.

(183) In addition, in the case illustrated in FIG. **9B**, since there is no object (user) that appears in common in the upper image and the lower image, it is not possible to determine the anteroposterior relationship using parallax. Therefore, when the face image is present only in the lower image, the gate device **10** may determine the anteroposterior relationship of the user according to the position of the user in the image (lower image).

(184) Alternatively, by combining the first and second example embodiments, an authentication target person may be selected using parallax in the case illustrated in FIGS. **8A** and **9A**, and an authentication target person may be selected according to the position of the image in the case illustrated in FIG. **9B**.

(185) The gate device **10** may change the control of the LED for dimming according to the purpose of the image (face image; biometric information) to be acquired. For example, the gate device **10** may change the light intensity between the case of imaging the user located in the authentication area and the case of imaging the entrant into the own device.

(186) A plurality of gate devices **10** may operate in conjunction with each other or in cooperation with each other. Specifically, the server device **20** may transmit a response to the authentication request acquired from one gate device to another gate device **10**. For example, in the example of FIG. **2**, the authentication result for the authentication request acquired from the gate device **10-1** may be transmitted to each of the gate devices **10-1** to **10-3**. In this case, even if the authentication target person detected by the gate device **10-1** moves toward the gate device **10-2** (even if the authentication target person enters the gate device **10-2**), the authentication target person can pass through the gate device **10-2**. In this case, the biometric information of the authentication target person generated by the gate device **10-1** is transmitted from the gate device **10-1** to the other gate devices **10-2** and **10-3**. Alternatively, the biometric information of the authentication target person may be transmitted from the server device **20** to the other gate devices **10-2** and **10-3**.

(187) The gate device **10** may be installed in a limited space of an existing facility, for example. In this case, the length of the main body of the gate device **10** in the traveling direction of the person who passes through the gate device **10** can be about 1.5 meters, which is shorter than 2 to 3 meters, which is the length of a general face authentication gate, and the distance to the gate **17** can be, for example, 96 centimeters (corresponding to one and a half steps of the stride of an adult male). When the length of the main body of the gate device **10** is about 1.5 meters, the length of the authentication area in the traveling direction is preferably about 30 cm to 60 cm.

(188) A form of data transmission and reception between the gate device **10** and the server device **20** is not particularly limited, but data transmitted and received between these devices may be encrypted. The face image and the feature amount calculated from the face image are personal information, and in order to appropriately protect the personal information, it is desirable that encrypted data be transmitted and received.

(189) In the flowchart (Sequence Diagram) used in the above description, a plurality of steps (processes) is described in order, but the execution order of the steps executed in the example embodiments is not limited to the described order. In the example embodiments, for example, the order of the illustrated steps can be changed within a range in which there is no problem in terms of content, such as executing each process in parallel.

(190) The above example embodiments have been described in detail in order to facilitate understanding of the present disclosure, and it is not intended that all the configurations described above are necessary. In addition, in a case where a plurality of example embodiments has been described, the example embodiments may be used alone or in combination. For example, a part of the configuration of the example embodiment can be replaced with the configuration of another example embodiment, or the configuration of another example embodiment can be added to the

configuration of the example embodiment. Further, it is possible to add, delete, and replace other configurations for a part of the configuration of the example embodiment.

(191) Although the industrial applicability of the present disclosure is apparent from the above description, the present disclosure can be suitably applied to an authentication system installed in an airport, a station, or the like.

(192) Some or all of the above example embodiments may be described as the following supplementary notes, but are not limited to the following.

(193) [Supplementary Note 1]

(194) A gate device including: an acquisition unit configured to acquire an image from a camera; a setting unit configured, in a case where a plurality of users appears in the acquired image, to specify a user in a foremost row among the plurality of users according to a position of each of the plurality of users in the image, and set the specified user as an authentication target person; and a gate control unit configured to control a gate based on an authentication result of the authentication target person.

[Supplementary Note 2]

(195) The gate device according to Supplementary Note 1, in which the camera is installed at a predetermined angle with respect to a traveling direction of the user.

(196) [Supplementary Note 3]

(197) The gate device according to Supplementary Note 1 or 2, in which the setting unit specifies a user located at an end on a side determined according to the traveling direction of the user and an installation position of the camera as the user in the foremost row.

(198) [Supplementary Note 4]

(199) The gate device according to any one of Supplementary Notes 1 to 3, in which the setting unit excludes a user whose interocular distance is smaller than a predetermined value from the determination related to the user in the foremost row among the plurality of users.

(200) [Supplementary Note 5]

(201) The gate device according to any one of Supplementary Notes 1 to 4, in which in a case where the camera is installed on a right side with respect to the traveling direction of the user, the setting unit specifies a user appearing at a right end of the image as the user in the foremost row.

(202) [Supplementary Note 6]

(203) The gate device according to any one of Supplementary Notes 1 to 4, in which in a case where the camera is installed on a left side with respect to the traveling direction of the user, the setting unit specifies a user appearing at a left end of the image as the user in the foremost row.

(204) [Supplementary Note 7]

(205) The gate device according to any one of Supplementary Notes 1 to 6, further including an authentication request unit configured to transmit an authentication request including biometric information of the authentication target person to a server device.

(206) [Supplementary Note 8]

(207) The gate device according to Supplementary Note 7, in which the authentication request unit transmits the authentication request including a feature amount generated from a face image of the authentication target person to the server device.

(208) [Supplementary Note 9]

(209) An authentication system including: a server device configured to store biometric information of each of a plurality of system users and execute authentication processing using the plurality of pieces of biometric information; and a gate device connected to the server device, in which the gate device includes: an acquisition unit configured to acquire an image from a camera; a setting unit that configured, in a case where a plurality of users appears in the acquired image, to specify a user in a foremost row among the plurality of users according to a position of each of the plurality of users in the image, and set the specified user as an authentication target person; an authentication request unit configured to request a server device to authenticate the authentication target person;

and a gate control unit configured to control a gate based on an authentication result of the authentication target person.

[Supplementary Note 10]

(210) The authentication system according to Supplementary Note 9, in which the authentication request unit transmits an authentication request including biometric information of the authentication target person to the server device, and the server device authenticates the authentication target person using the biometric information acquired from the authentication request and the biometric information of the plurality of system users.

[Supplementary Note 11] A gate device control method including: in a gate device, acquiring an image from a camera; in a case where a plurality of users appears in the acquired image, specifying a user in a foremost row among the plurality of users according to a position of each of the plurality of users in the image, and setting the specified user as an authentication target person; and controlling a gate based on an authentication result of the authentication target person.

[Supplementary Note 12]

(211) A computer-readable storage medium storing a program for causing a computer mounted in a gate device to execute: acquiring an image from a camera; in a case where a plurality of users appears in the acquired image, specifying a user in a foremost row among the plurality of users according to a position of each of the plurality of users in the image, and setting the specified user as an authentication target person; and controlling a gate based on an authentication result of the authentication target person.

(212) The disclosures of the cited prior art documents are incorporated herein by reference. While the example embodiments of the present disclosure have been particularly shown and described above, the present disclosure is not limited to these example embodiments. It will be understood by those skilled in the art that these example embodiments are exemplary only and that various modifications may be made without departing from the scope and spirit of the present disclosure as defined by the claims. That is, it goes without saying that the present disclosure includes various modifications and corrections that can be made by those skilled in the art in accordance with the entire disclosure including the claims and the technical idea.

## REFERENCE SIGNS LIST

(213) **10**, **10-1** to **10-3**, **100** gate device **11** face authentication module **12-1** upper camera **12-2** lower camera **13** display **14** display **15** area sensor **16** in-gate sensor **17** gate **20** server device **101** acquisition unit **102** setting unit **103**, **207** gate control unit **201**, **301** communication control unit **202** user detection unit **203** biometric information acquisition unit **204** authentication target setting unit **205** authentication request unit **206** entrant determination unit **208** message output unit **209**, **304** storage unit **302** user registration unit **303** authentication unit **311** processor **312** memory **313** input/output interface **314** communication interface

## Claims

1. A gate device comprising: a memory storing instructions; and at least one processor configured to execute the instructions to perform: acquiring a first image from a first camera and a second image from a second camera, the first image and the second image having a common region; in a case where heads of a plurality of users simultaneously appear in the common region of the first image and the second image, specifying a user in a foremost row among the plurality of users according to a position of each of the plurality of users in the common region of the first image and the second image, and setting the specified user as an authentication target person; detecting an entrant into a gate; determining whether the entrant and the authentication target person are the same person; controlling the gate based on an authentication result of the authentication target person and whether the entrant and the authentication target person are the same person; and in a case where the plurality of users appear in only the first image or in only the second image, setting

an error result, wherein the specified user is set as the authentication target person by: setting a distance measurement band in the common region; dividing the distance measurement band into a plurality of areas; calculating, for each area, a distance to a camera installation surface of the first and second cameras, using a stereo method; specifying a position where each of a plurality of distance peaks appears, a number of the plurality of distance peaks corresponding to a number of the plurality of users; and specifying, as the user in the foremost row, a user corresponding to a position where a minimum distance peak appears.

2. The gate device according to claim 1, wherein the first camera the second camera are each installed at a predetermined angle with respect to a traveling direction of the plurality of users.

3. The gate device according to claim 1, wherein the at least one processor is configured to execute the instructions to perform: specifying the user in the foremost row from the plurality of users located at an end on a side determined according to a traveling direction of the plurality of users and an installation position of the first camera and the second camera.

4. The gate device according to claim 1, wherein the at least one processor is configured to execute the instructions to perform: excluding any user having an interocular distance smaller than a predetermined value from being specified as the user in the foremost row.

5. The gate device according to claim 1, wherein the at least one processor is configured to execute the instructions to further perform: in a case where the first camera and the second camera are installed on a right side with respect to a traveling direction of the plurality of users, specifying the user in the foremost row from the plurality of users appearing at a right end of the common region of the first image and the second image.

6. The gate device according to claim 1, wherein the at least one processor is configured to execute the instructions to perform: in a case where the first camera and the second camera are installed on a left side with respect to a traveling direction of the plurality of users, specifying the user in the foremost row from the plurality of users appearing at a left end of the common region of the first image and the second image.

7. The gate device according to claim 1, wherein the at least one processor is further configured to execute the instructions to perform: transmitting an authentication request including biometric information of the authentication target person to a server device.

8. The gate device according to claim 7, wherein the at least one processor is configured to execute the instructions to perform: transmitting the authentication request including a feature amount generated from a face image of the authentication target person to the server device.

9. A gate device control method performed by a gate device and comprising: acquiring a first image from a first camera and a second image from a second camera, the first image and the second image having a common region; in a case where heads of a plurality of users simultaneously appear in the common region of the first image and the second image, specifying a user in a foremost row among the plurality of users according to a position of each of the plurality of users in the common region of the first image and the second image, and setting the specified user as an authentication target person; detecting an entrant into a gate; determining whether the entrant and the authentication target person are the same person; controlling the gate based on an authentication result of the authentication target person and whether the entrant and the authentication target person are the same person; and in a case where the plurality of users appear in only the first image or in only the second image, setting an error result, wherein the specified user is set as the authentication target person by: setting a distance measurement band in the common region; dividing the distance measurement band into a plurality of areas; calculating, for each area, a distance to a camera installation surface of the first and second cameras, using a stereo method; specifying a position where each of a plurality of distance peaks appears, a number of the plurality of distance peaks corresponding to a number of the plurality of users; and specifying, as the user in the foremost row, a user corresponding to a position where a minimum distance peak appears.

10. A non-transitory computer-readable storage medium storing a program for causing a computer

mounted in a gate device to execute: acquiring a first image from a first camera and a second image from a second camera, the first image and the second image having a common region; in a case where heads of a plurality of users simultaneously appear in the common region of the first image and the second image, specifying a user in a foremost row among the plurality of users according to a position of each of the plurality of users in the common region of the first image and the second image, and setting the specified user as an authentication target person; detecting an entrant into a gate; determining whether the entrant and the authentication target person are the same person; controlling the gate based on an authentication result of the authentication target person and whether the entrant and the authentication target person are the same person; and in a case where the plurality of users appear in only the first image or in only the second image, setting an error result, wherein the specified user is set as the authentication target person by: setting a distance measurement band in the common region; dividing the distance measurement band into a plurality of areas; calculating, for each area, a distance to a camera installation surface of the first and second cameras, using a stereo method; specifying a position where each of a plurality of distance peaks appears, a number of the plurality of distance peaks corresponding to a number of the plurality of users; and specifying, as the user in the foremost row, a user corresponding to a position where a minimum distance peak appears.

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